

Jyot Antani, PhD

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Expertise: Biophysics, Microbiology, Virology, Optical Microscopy, Biotechnology, Bacteriophage Therapy

Education

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| 2022 | Optical Microscopy & Imaging in the Biomedical Sciences (OMIBS), Woods Hole (US)
Highly selective advanced light microscopy course at the Marine Biological Laboratory |
| 2021 | Doctor of Philosophy (Ph.D.), Department of Chemical Engineering, Texas A&M University (US)
Dissertation: Sensory Functions of the Bacterial Flagellar Motor |
| 2016- | Bachelor of Technology with Honours in Chemical Engineering
Indian Institute of Technology Bombay (IIT Bombay), Mumbai (India) |

Key Research Experience

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| Jun 2021-
Now | <p>Yale University, New Haven, CT (US)
Associate Research Scientist (2025-Now); Postdoctoral Associate (Jun 2021 - Mar 2025)
Paul Turner Lab. Department of Ecology and Evolutionary Biology; Quantitative Biology Institute; Center for Phage Biology & Therapy</p> <ul style="list-style-type: none"> Founded and led the Center's biophysics/visualization effort; developed single-particle microscopy to measure interactions between individual viral particles and host cells (Antani^{et al.} 2024 <i>PNAS</i>, Patent) Conducted experimental evolution to determine how single-cell motility trait distributions shift under selection by a flagellum-binding phage (Antani^{et al.}, <i>bioRxiv</i>, In review at <i>PLOS Pathogens</i>) Advanced evolution-informed phage therapy strategies targeting multi-drug-resistant bacterial infections (Oromí-Bosch*, Antani*, & Turner 2023 <i>Annu Rev Virol</i>; Antani^{et al.} & Turner 2026 <i>npj Viruses</i>; Sikkema et al. 2026 <i>PNAS</i>; ...) * equal contribution |
| Aug 2016-
May 2021 | <p>Texas A&M University, College Station, TX (US)
Pushkar Lele Lab, Department of Chemical Engineering</p> <ul style="list-style-type: none"> Determined how mechanical load/force regulates chemotactic signaling at the bacterial flagellar motor using phase-contrast, dark-field, and TIRF imaging and optical tweezers (Antani et al. 2021 <i>Nat Commun</i>) Developed a microscopy-based assay to measure chemotactic activity and swimming statistics in <i>H. pylori</i>, the gastric carcinogen (Antani et al. 2021 <i>eLife</i>) Design and development of experimental and analytical framework (Katiyar*, Antani* et al. 2021 <i>J Microsc</i>) |

Academic Appointments

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| 2025-2028 | Editor, mSystems – American Society for Microbiology (ASM) journal on microbial systems biology |
| 2023-Now | Research facilitator (instructor) , Optical Microscopy summer course , Woods Hole, MA (US) |

Patent

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| | Turner PE, Antani JD, Emonet T. Methods of high-throughput quantification of virus particle attachment. Yale University Patent Application (2025) PCT/US25/49613. |
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Invited Talks/Seminars

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| Jun 2026 | American Society for Microbiology Annual Microbe Meeting 2026, Washington DC (US). <i>Advanced Optical Microscopy Approaches for Quantifying Microbial Spatial Organization and Dynamics</i> . |
| Jan 2026 | American Society for Microbiology Connecticut Valley Branch Meeting. <i>Multi-dimensional Measurements of Phage-Bacteria Interactions</i> |
| Nov 2023 | Departmental Seminar, Chemical Engineering, Indian Institute of Technology Bombay (India). Microbial Treasure Hunt: How Bacteria and Viruses Find their Haven . |
| Oct 2021 | Howard Berg Lab, Harvard U (US). <i>Force-regulation of CheY-P-binding to the flagellar motor</i> . |

Antani JD✉, Ward T, Emonet T, Turner PE (2024) High-throughput quantification of virus particle attachment to host bacterial cells. *PNAS, USA*. doi: [10.1073/pnas.2410905121](#)

Antani JD✉, Theroux A, Avgidis F, Scharf B, Emonet T, Turner PE. Flagellotropic bacteriophage selects for evolutionary remodeling of cell motility and chemotaxis in *Escherichia coli*. *bioRxiv*. doi:[10.1101/2025.05.06.652435](#)
In review at *PLOS Pathogens*.

Oromi-Bosch A*, Antani JD*, Turner PE (2023) Developing phage therapy that overcomes the evolution of bacterial resistance. *Annu Rev Virology*. doi: [10.1146/annurev-virology-012423-110530](#)

Antani JD✉, Turner PE (2026) Rage Against the Mean: A Perspective on Measuring Fitness of Individual Phage Particles. *npj Viruses*. doi:[10.1038/s44298-026-00187-4](#)

Antani JD, Gupta R, Lee AH, Rhee KY, Manson MD, Lele PP (2021) Mechanosensitive recruitment of stator units promotes binding of the response regulator CheY-P to the flagellar motor. *Nature Communications*. doi: [10.1038/s41467-021-25774-2](#)

Antani JD, Sumali AX, Lele TP, Lele PP (2021) Asymmetric random walks reveal that the chemotaxis network modulates flagellar rotational bias in *Helicobacter pylori*. *eLife*. doi: [10.7554/eLife.63936](#)
This work helped secure an R01 research grant totaling over \$1.3 million from the National Institute of General Medical Sciences, National Institute of Health. <https://www.eurekalert.org/news-releases/945164>

Sikkema A, ..., Antani JD et al. (2026) A fully synthetic Golden Gate Assembly system for engineering a *Pseudomonas aeruginosa* phiKMV-like phage. *PNAS, USA*. doi:[10.1073/pnas.2525963123](#)
A collaborative synthetic biology project with New England Biolabs, a leading biotechnology company.

Katiyar A*, Antani JD*, McKee BP, Gupta R, Lele PP, Lele TP (2021) A method for direct imaging of x-z cross-sections of fluorescent samples. *Journal of Microscopy*. doi: [10.1111/jmi.12965](#)

Wong GCL, Antani JD, et al. (2021) Roadmap on emerging concepts in the physical biology of bacterial biofilms: from surface sensing to community formation. *Physical Biology*. doi: [10.1088/1478-3975/abdc0e](#)

Hernandez CA, ..., Antani JD, Turner PE (2026) Prophage activity shapes the thermal ecology and evolution of a marine bacterial host. *bioRxiv*. doi: [10.64898/2026.02.19.706829](#). In revision at *PNAS*.

Nasserredine A,..., Antani JD,..., King MC, Horsley V (2026) SUN2 mediates epigenetic remodeling to drive mechanotransduction during skin fibrosis. *bioRxiv*. doi: [10.64898/2026.03.19.712957](#). In revision, *J Clin Invest*.

Antani JD, Shaji A, Gupta R, Lele PP (2024) Reassessing the standard chemotaxis framework for understanding biased migration in *Helicobacter pylori*. *Annu Rev Chem Biomol Eng*. doi: [10.1146/annurev-chembioeng-100722-114625](#)

Katiyar A, Zhang J, Antani JD,..., Lele TP (2022) The Nucleus Bypasses Obstacles by Deforming Like a Drop with Surface Tension Mediated by Lamin A/C. *Advanced Science*. doi: [10.1002/adv.202201248](#)

Ford KM, Antani JD, Nagarajan A, Johnson MM, Lele PP (2018) Switching and torque generation in swarming cells of *E. coli*. *Frontiers in Microbiology*, 9, 2197. doi: [10.3389/fmicb.2018.02197](#)

Pre-prints (published, yet to be peer-reviewed)

Müller DM,..., Antani JD,..., Bollyky PL (2024) Bacterial receptors but not anti-phage defence mechanisms determine host range for a pair of *Pseudomonas aeruginosa* Lytic Phages. *bioRxiv*. doi: [10.1101/2024.04.30.591980v2](#)

Kanhirun N,..., Antani JD,..., Pride DT (2024) Identification of a large cohort of *Enterobacter jumbo* phages with broad host ranges across pathogenic Gammaproteobacteria. *bioRxiv*. doi: [10.1101/2024.10.29.620934](#)

Publications in preparation

Targ B, Turner PE, Antani JD✉. CPFUcounter: An open-source app for counting bacterial colonies/viral plaques.

Antani JD✉, Williams A, An A, Turner PE. Multiscale measurements of interactions between jumbo phages and *Pseudomonas aeruginosa* patient isolates. *Presented at IAS-USA Conference on Bacteriophages, Washington DC, Oct 2025*

Davidson M, ..., Antani JD, et al. Phage selection of *Salmonella* Typhimurium results in cellulose synthesis mutants. Presented at American Society of Microbiology's 2025 annual meeting in Los Angeles, CA (USA).

Arenas-Castro H, Antani JD, et al. Patterns of post-pollination prezygotic isolation in monkeyflowers.

Selected Peer Reviews (Independent, 80+ total)

[Full list on WebOfScience \(Link\)](#)

Microbiology Spectrum (22)
Nature Microbiology (1)

Physical Biology (4)
PLoS One (4)

Applied & Environmental Microbiology (7)

Honors & Awards

2026	Yale's nominee for Blavatnik Regional Award
2025	Peggy Cotter Award , American Society for Microbiology
2025	New Investigator Scholarship, International Antiviral Society–USA Conference on Bacteriophages: Biology, Dynamics, and Therapeutics
2023	Top Reviewer 2023, <i>Microbiology Spectrum</i> – an American Society for Microbiology (ASM) Journal
2023	Best Poster Presentation Award, Yale Postdoctoral Association Annual Symposium 2023
2022	MBL Alumni ROCS (Regional Outreach and Communication in STEM) Grant
2022	BioImaging North America (BINA) Professional Development Award - Chan Zuckerberg Initiative
2022	Histochemical Society (HCS) Travel Award
2020	CIRTL Associate Certificate, Center for the Integration of Research, Teaching, and Learning
2015	Best Poster Award, AMAT seminar , Mumbai (India)
2015	Undergraduate Research Award URA01, IIT Bombay
2012	All India Rank 1427 in Joint Entrance Examination for engineering, among 479,651 participants
2008	National Talent Search Scholarship, NCERT, India

Grant-Writing (Submitted, pending)

Experience writing diverse multi-disciplinary grants with variable focus on imaging, for instance:

NIH R01	Stephen I. Katz Early Stage Investigator Research Project Grant (NIBIB / NIAID): <i>A Generalizable Single-Particle Platform for Viral Entry Biophysics and Phage-based Gene Delivery in Mammalian Cells</i>
NIH R21	Exploratory/Developmental Bioengineering Research Grants (NCI): <i>Programmable Synthetic Bacteriophage Vehicles for Targeted Cancer Therapeutic Cargo Delivery</i>
DARPA	Biological Technologies Office: <i>High-Throughput Single-Virion Entry-to-Fate Microscopy for Rapid Host/Pathogen Compatibility Profiling</i>
NSF	DBIO Evolutionary Processes: <i>Convergence and constraints in the evolution of bacterial interactions with flagella-specific phages</i>
DOE	Genesis mission: <i>Predicting Phage-Bacteria Interactions Across Biological Scales Using Generative AI</i>
Foundation	Alfred P Sloan Foundation: <i>Adaptive evolution under predation as a quantifiable signature of life: convergence and constraints in bacterial responses to motility-targeting phages</i>

Key Conference Presentations

Antani JD, Williams A, An A, Turner PE. Jumbo Phages effectively kill patient isolates despite slow attachment.

Oct 2025	Conference on Bacteriophages, International Antiviral Society, Washington, DC (US)
Jun 2025	Microbe- Annual meeting of the American Society for Microbiology, Los Angeles (US)
Apr 2025	American Society for Microbiology Joint Branch Meeting, New England & Connecticut Valley Branches (US)
Apr 2025	Emerging Pathogens Initiative Meeting, Howard Hughes Medical Institute (US)

Antani JD, Ward TJ, Emonet T, Turner PE. Quantitative biology of bacterial virus attachment to host surfaces.

Mar 2024	American Physical Society March Meeting 2024 , Minneapolis (US)
Jan 2024	Bacteria versus Phage: the Main Event , Princeton Center for Theoretical Science Workshop, Princeton University (US)
Jun 2023	Microbe- Annual meeting of the American Society for Microbiology, Houston (US)
May 2023	Yale Postdoctoral Association Symposium, Yale University, New Haven (US)
Sep 2022	Annual Microbiology Retreat, Yale University, New Haven (US)

Jun 2022 | [Microbe- Annual meeting of the American Society for Microbiology, Washington, DC \(US\)](#)

Antani JD, Theroux A, Emonet T, Turner PE. Bacterial Evolution of Resistance To A Flagellotropic Phage.

Sep 2023 | Annual Microbiology Retreat, Yale University, New Haven (US)

Jun 2023 | [Microbe- Annual meeting of the American Society for Microbiology, Houston \(US\)](#)

Jun 2022 | [Microbe- Annual meeting of the American Society for Microbiology, Washington, DC \(US\)](#)

Antani JD, Lele PP. Modulation of chemotactic signaling in bacteria by mechanical forces.

Mar 2019 | ChEGSA Annual Symposium, Texas A&M University, College Station (US)

Oct 2018 | [American Institute of Chemical Engineers \(AIChE\) Annual Meeting, Pittsburgh \(US\)](#)

Dec 2017 | Texas Branch of American Society for Microbiology, 2017 Fall Meeting, College Station (US).

Antani JD, Lele PP. Asymmetric Swimming and Chemotaxis in *Helicobacter pylori*.

Jan 2021 | [Bacterial Locomotion And Signal Transduction \(BLAST\) XVI Meeting, Zoom \(online\)](#)

Feb 2020 | [Biophysical Society \(BPS\) Annual Meeting, San Diego \(US\)](#)

Jan 2019 | [Bacterial Locomotion And Signal Transduction \(BLAST\) XV Meeting, New Orleans \(US\)](#)

Antani JD, Venkatesh KV. Analysis of Glucose Homeostasis in Humans.

Nov 2015 | [Futuristic Approach for Alternatives to Animal Testing, CEFIPRA seminar, Mumbai \(India\)](#)

Media mentions

- [Times of India – Amdavadi scientist at Yale aims to combat AMR bacteria using viruses](#)
- [YaleNews – New tool offers insights into virus-cell interactions](#)
- [Phys.org – Fluorescence microscopy tracks phage attachment to bacteria in real time](#)
- [ASM Microcosm magazine – Friends and Foes: Microbial Interactions and Infection](#)
- [Texas A&M Stories – To Colonize Different Environments, Bacteria Precisely Tune Their Nanomotors](#)
- [Phys.org – A novel approach to determine how carcinogenic bacteria find their targets](#)
- [Texas A&M Stories – A Novel Approach To Determine How Carcinogenic Bacteria Find Their Targets](#)
- 2000-word profile feature in *Divya Bhaskar*, a leading Gujarati-language newspaper, highlighting scientific career following an in-depth interview.

Technical Skills

Bacteria, Mammalian cell, Virus Culturing	Assay Development	Optical Microscopy: advanced expertise
Perfusion & Thermal Control Experiments	Microfluidics Experiments	Image Analysis Automation
Genome Analysis/Bioinformatics	Prototyping & 3D-printing	Single Particle Tracking
Electron Microscopy (Negative-stain)	qPCR, Genome editing	

Computational Skills

Image Processing & Analysis:	MATLAB, Python, ImageJ – {Highlights: particle-tracking; segmentation}
Genetic Analysis:	Molecular Biology {SnapGene, Benchling}, Bioinformatics
Design & Development:	SolidWorks, AutoCAD, LabVIEW,
Simulation Packages:	ANSYS Fluent, Aspen Plus, OpenFOAM, COMSOL, CHARMM, VMD
Programming:	MATLAB, Python, C, C++, Fortran, HTML

Teaching & Mentoring Experience

Guest Lecturer

Fall 2026 | Quinnipiac University, Virology Course. Instructor of record: [Nicole Bracci, Ph.D.](#)

Teaching Assistant at Texas A&M Chemical Engineering

Spring 2019 | Heat Transfer Operations

Spring 2017 | Bioprocess Engineering

Fall 2016 | Bioprocess Engineering

Workshop

Fall 2019 | A hands-on workshop for image processing and single particle tracking in MATLAB
Pushkar Lele Lab, Texas A&M University

Graduate students mentored

Jan 2025-
Now | [Gunalan Natesan](#) & [Emma Keeler](#), Graduate Students, Yale University
Imaging-based approaches to measure mammalian virus titers, bacterial lineages, and phage-bacteria interactions

Undergraduate students mentored

Sep 2024-
Now | [Alex Williams](#) & [Allan An](#), Undergraduate Researchers, Yale University
Interactions between jumbo phages and clinical isolates of *Pseudomonas aeruginosa*

June 2025-
Now | [Bobby Targ](#), Undergraduate Intern, Yale University
Bioinformatics pipelines supporting microbiology assays

Aug 2022-
May 2023 | [Timothy Ward](#), Undergraduate Researcher, Yale University
Classical bacteriophage adsorption assays. Co-author on [Antani et al. 2024 PNAS](#)
Best Poster Award winner at [Yale Undergraduate Research Symposium](#), March, 2023

Oct 2021-
Oct 2022 | [Austen Theroux](#), STARS II Science & Quantitative Reasoning Education Scholar, Yale University
Evolution of motile bacteria against a bacteriophage that attaches through the flagellum
Presented a poster about this work at *Microbe* ASM meeting at Washington DC in June, 2022

Dec 2019-
Jan 2021 | [Anita Sumali](#), Gathright Scholar, Biomedical Engineering, Texas A&M University
Quantitative analysis of the run-reversal motility of *H. pylori*; co-author on [Antani et al. 2021 eLife](#)
Currently enrolled in EnMed (MEng/MD dual degree program) at Texas Medical Center, Houston (US)

Apr 2017-
Apr 2018 | AggIE_Challenge – Undergraduate Mentoring, Texas A&M University
Graduate-mentor to a multidisciplinary team of eight undergraduate students employing 3D-printing technology to design scaled-up bacterial models, in order to study the hydrodynamic effects of cell morphology on motility

Teaching, Mentoring, & Management Training

Sep 2021-
Dec 2021 | Auburn University's Preparing Future Faculty Workshop
Interactive workshops consisting panel discussions from faculty and university administrators

Sep 2021 | [Fundamentals of Inclusive Teaching](#)
[Poorvu Center: Graduate and Postdoctoral Teaching Development, Yale University](#)
A 4-part workshop on creating a respectful, supportive, equitable, and transparent classroom environment

Aug 2018-
Dec 2019 | [Academy for Future Faculty \(AFF\)](#), Center for Teaching Excellence, Texas A&M University
Faculty Mentor: Dr. Micah Green, Texas A&M University
A program designed for training graduate students as future faculty members, featuring weekly seminars and workshops, and anchored by faculty mentorship

Jul-
Aug 2019 | An Introduction to Evidence-Based Undergraduate STEM Teaching
Muti-university Massive Open Online Course from the [NSF-founded CIRTl network](#)
Eight-week online course on [edX.org](#) catering introduction to effective teaching strategies and the research that supports them. Attended local learning community meetings for the course at Texas A&M University.

Oct 2019 | Improving Research Mentoring Relationships – Introduction to Effective Mentorship
Center for the Improvement of Mentored Experiences in Research (CIMER)
1-day workshop at Texas A&M University, conducted by CIMER from University of Wisconsin-Madison

Service, Leadership, & Management Experience

Jul 2025- Now	Councilor, American Society for Microbiology Connecticut Valley branch - Organizing, coordinating, and budgeting for microbiology activities (conferences, awards disbursement) serving scientists in Connecticut, Rhode Island, parts of New Hampshire, and parts of Massachusetts - Representing the local branch in strategic meetings at national ASM organization
Jul 2024- Now	Communications Committee Member, American Society for Microbiology Member of a team making outreach efforts for the organization ASM with more than 32,000 members worldwide
Jan 2022- Jun 2024	Communications Committee Manager and Coordinator, Yale Postdoctoral Association (YPA) - Led and coordinated cross-functional teams (communications, operations) to support 1000+ researchers - Organized a drive for T-shirts and jackets benefiting 500+ postdoctoral scientists across university
Oct 2018- Aug 2020	Secretary (2018-19) & Chair (2019-20), Chemical Engineering Graduate Student Advisory Council <i>Artie McFerrin Department of Chemical Engineering, Texas A&M University</i> - Designed the council by-laws as a founding member - Set up channels of communication through surveys and anonymous feedback mechanisms, between grad students and administration in the department - Addressed grad student welfare issues including salary, benefits, mental well-being, work-life balance, student-advisor relationships, facilities, career events, and social events
Apr 2014- Mar 2015	Competitions Manager, AZeotropy 2015 <i>IIT Bombay's Intercollegiate Chemical Engineering Symposium</i> - Pioneered a new model-making competition <i>ChemETimer</i> based on time-precision - Designed a presentation-based competition <i>Swaasthya</i> on engineering-applications to healthcare - Designed <i>Cipher</i>, the online cryptic hunt which received a participation of 400+ - Actualized unconventional chemical engineering quizzing for <i>Chem-O-Philia</i>, national level quiz

Science Outreach

May 2025	Glowing Viruses: How Strong is Their Grip? Wrote a summary article explaining the PNAS 2024 work on visualizing virus attachment to bacteria
May 2024	Flipped Science Fair at Yale University Presented a poster about current research topic to middle school students in the New Haven County, CT
Mar 2023	Tiny World for Little Humans: Teaching Quantitative Microscopy to Underserved Youth - Taught quantitative microscopy through the use of paper microscopes (foldscope.com) to middle school students at Fair Haven School, New Haven, Connecticut - Outreach project funded by the MBL ROCS- Marine Biological Laboratory Regional Outreach and Communication in STEM
Feb 2022- Sep 2022	Associate Writer & Editor, LabLeaks A biweekly mailing list summarizing recent noteworthy scientific articles to the general audience.
Feb 2022	Exploring Science for Middle School Students Presented doctoral research topic to middle school children in the New Haven County, Connecticut
Jul 2020	Coronasurveys: Monitoring the Incidence of COVID-19 via Open Surveys Volunteered services as the primary translator of the survey to the Gujarati language.